Sandeep Singh R

Bengaluru, Karnataka | Ph: 7406050506 | Email | LinkedIn | Github | Portfolio

PROFESSIONAL SUMMARY

PMP-certified Project Manager and XLRI-certified Product Manager with 10+ years of cross-functional experience spanning Embedded Systems, IoT, Semiconductor, and Automation domains. Proven track record in full project lifecycle delivery, Agile/Scrum execution, stakeholder management, and product roadmap ownership. Adept at leading engineering teams, managing budgets and risk, and driving on-time delivery with measurable quality outcomes. Equipped with a solid foundation in technical product thinking — including PRD/user-story definition and GTM alignment — making a natural fit for Junior Product Manager opportunities alongside senior Project Manager roles. Experienced in regulatory compliance (ISO, DO-178, MISRA C) and certified in both project and product disciplines.

CORE COMPETENCIES

Project Management	Agile & Scrum, PMP, PDLC, Program Management, Risk Register, Change Control, Milestone Tracking, Scope & Schedule Management, Stakeholder Management, Budgeting & P&L, Resource Allocation, Vendor Negotiation
Product Management	Product Roadmap, User Stories, PRD/MRD, Feature Prioritization, OKRs/KPIs, Go-to-Market (GTM), Customer Discovery, Voice of Customer (VoC), Competitive Analysis, Product-Market Fit
Technical Expertise	Embedded Systems, IoT, Firmware Development, Embedded C / C, Python, MATLAB, FreeRTOS, STM32, PCB Design (KiCad, Altium)
Process & Quality	Workflow Automation, QA & Validation, ISO / DO-178 / MISRA C Compliance, Design Reviews, Technical Documentation, CI/CD, Version Control
Tools & Platforms	JIRA, Bitbucket, Git, MATLAB GUI, Jython, Agile Boards, Sprint Planning
Leadership	Team Building, Cross-Functional Collaboration, Mentoring, Workshop Facilitation, Client Engagement

PROFESSIONAL EXPERIENCE

Project Manager | *Klug Avalon Mechatronics Pvt. Ltd.* *Jun 2023 – Present*

- Reduced crane downtime incidents by 40% by leading an 8-member engineering team in design, deployment and calibration of Safe Load Indicator (SLI) systems — owning the full project scope, risk register, and milestone schedule.
- Achieved 100% on-time delivery across 5+ major projects through Agile sprint planning, resource allocation, and change-control discipline aligned to PMO standards.
- Drove client satisfaction to 95% by aligning stakeholder expectations with project goals through structured review cadences and clear deliverable sign-offs.
- Improved engineering workflow efficiency by 30% via cross-functional process optimization, reducing inter-team handoff delays and rework cycles.
- Maintained P&L oversight for project budgets, negotiating vendor contracts and optimizing component sourcing to reduce procurement costs.

Assistant Manager – R&D | *MPA Pvt. Ltd.* *May 2022 – Jun 2023*

- Cut development cycles by 20% by implementing Agile practices, sprint retrospectives, and embedded system optimization techniques.
- Boosted device performance by 25% through targeted firmware enhancements and efficient hardware resource utilization.
- Maintained 95% client satisfaction by acting as the bridge between product and engineering teams — defining requirements, managing scope, and aligning delivery timelines.
- Improved vendor coordination and component sourcing efficiency by 15% through structured RFQ processes and supplier SLAs.

Senior Embedded Design Engineer | *STPL* *Mar 2021 – May 2022*

- Improved system stability by 30% through systematic integration of hardware and software modules, supported by structured design reviews.
- Reduced product defects by 40% by introducing enhanced QA processes, formal design reviews, and structured validation frameworks.
- Contributed to product roadmap execution — translating customer requirements into embedded feature specifications and delivery milestones.

Career Break | Nov 2019 – Mar 2021

Personal bereavement (loss of father) coinciding with the COVID-19 pandemic. During this period, actively continued professional development as a freelance MATLAB and Embedded Systems Trainer/Developer — delivering online training sessions, developing client-specific MATLAB scripts and embedded prototypes.

Senior Software Developer (Client: Microchip Pvt. Ltd.) | *Nexsemi Systems Pvt. Ltd. Oct 2018 – Nov 2019*

Validation Engineer (Mar 2019 – Sep 2019)

- Improved testing efficiency by 80% by automating manual test cases using Jython scripting, building reusable validation frameworks.
- Enhanced bug detection coverage by 25% through structured test planning; reduced regression testing effort by 15%.

MATLAB Developer (Oct 2018 – Mar 2019)

- Increased visualization efficiency by 35% by developing advanced MATLAB-based GUI dashboards used for engineering decision-making.
- Prototyped embedded algorithms enabling faster simulation, performance testing, and accelerated release cycles.

Core Embedded Contributions (Nexsemi)

- Boosted product efficiency by 20% via critical embedded software updates; reduced deployment errors by 25% through CI/CD automation and version control discipline.

Embedded Design Engineer | *Monee & Company Pvt. Ltd.* *Nov 2016 – Nov 2017*

Company was subsequently shut down.

- Improved system stability by 20% designing and testing embedded hardware solutions; reduced prototyping timelines by 15% through PCB layout and firmware modularization.

Research Associate | *PESIT (Higher Education Pursuit)* *Oct 2012 – Aug 2014*

Role concluded to pursue M.Tech (Signal Processing) full-time at SJCIT, VTU.

- Enhanced data accuracy by 95% through research on acoustic systems and algorithm development.
- Published 5 National Conference Papers and 2 International Journal articles; mentored undergraduate research teams.

Design Engineer – Embedded | *Apsis Solutions* *Jun 2011 – Oct 2012*

- Improved embedded performance by 15%; reduced hardware-software integration issues by 20% through systematic debugging and documentation.

KEY PROJECTS

Safe Load Indicator (SLI) System for Cranes

- Reduced crane safety incidents by 40%; enhanced calibration processes cutting setup time by 25%.
- End-to-end ownership: stakeholder requirements → hardware/firmware specification → deployment → client sign-off.

Spectrometer Design & Development

- Developed STM32 firmware improving data acquisition accuracy by 30%; optimised testing methodologies reducing hardware integration time by 20%.

Automatic Flight Control System (AFCS)

- Implemented FreeRTOS integration improving system responsiveness by 15%; led hardware rig testing reducing system failures by 25%.

MATLAB GUI Tooling (Microchip / Freelance)

- Delivered dashboards and simulation tools that accelerated engineering decision-making and training workflows.

EDUCATION & CERTIFICATIONS

M.Tech – Signal Processing | SJCIT, VTU | 2016
B.E – Electronics & Communication | SSEC, VTU | 2011

Certifications

- PMP® – Project Management Professional (PMI)
- Product Management Certification – XLRI
- Embedded Systems & Peripherals – Microchip India Pvt. Ltd.
- Online Certifications – Project Management & Embedded Systems (Coursera)

PATENT

Automobile Emergency Alerting Autonomous Device – Design Patent Filed, India (Reg. No. 453281-001) | Mar 2025

ACHIEVEMENTS & ADDITIONAL

- Winner – Best Design, Embedded Design Contest (EFY)
- Published 5 National Conference Papers & 2 International Journals
- MATLAB Instructor – Udemy (Basic Course); conducted multiple engineering student workshops
- NCC 'B' Certificate Holder
- Passionate about adventure sports: rock climbing, rappelling, trekking

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